Computer Architecture & Concurrency – COMP0008
Lecture 2 Part 1: Abstracting the machine

Kevin Bryson
K.Bryson@ucl.ac.uk
Darwin Building Room 632.

Overview of the Lecture

- Lecture 1 was very much an overview of lots of ideas about computer architecture and concurrency concepts.
- In this lecture I am going to draw together a few key ideas from this previous lecture to begin looking at the architecture of a MIPS32 machine.

- This part will look at how a computer can be abstracted in terms of a von Neumann architecture and how this relates to real hardware.

EDVAC and von Neumann Architecture

First Draft of a Report
on the EDVAC

by

John von Neumann

Contract No. W–670–ORD–4926

Between the

United States Army Ordnance Department

and the

University of Pennsylvania

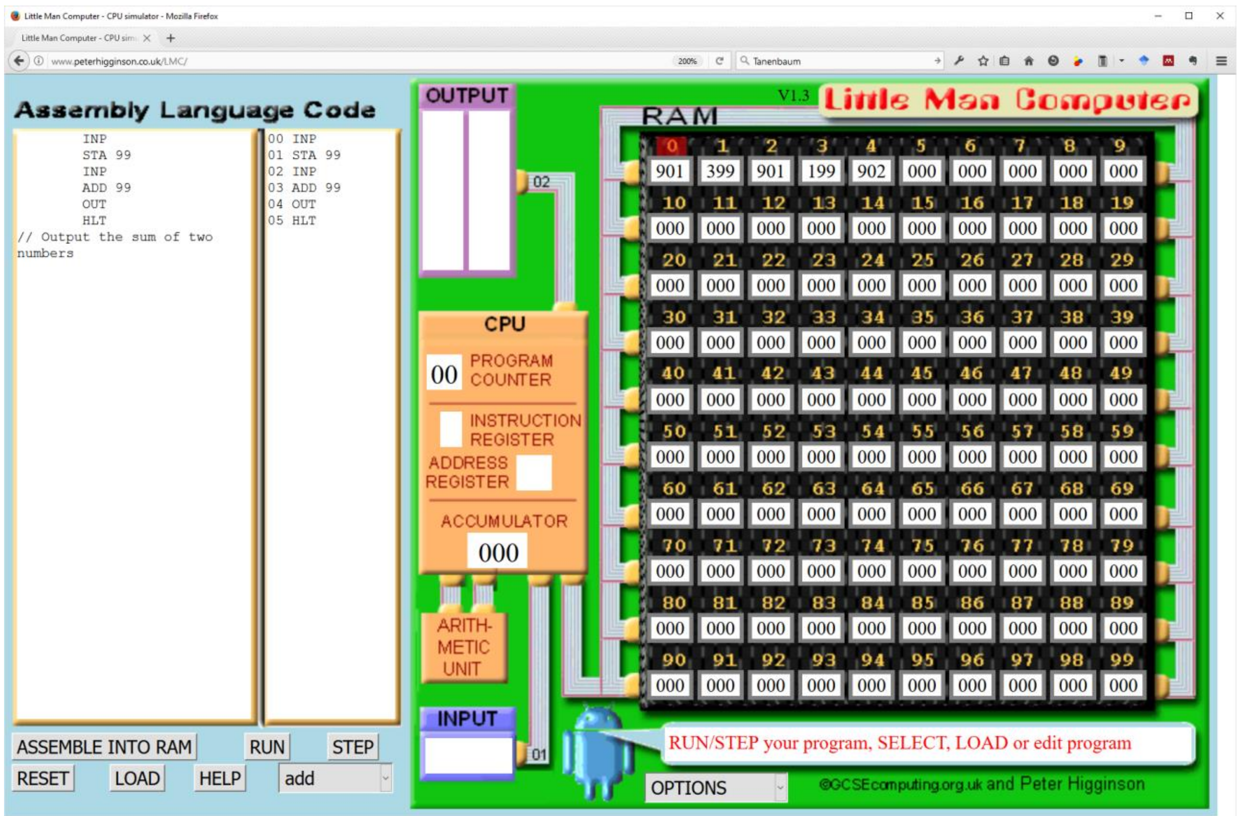
2.0 MAIN SUBDIVISIONS OF THE SYSTEM

An abstract analogy for a real computer …

- The *Little Man Computer* (LMC) was originally created by Dr. Stuart Madnick at MIT in 1965 as a simple analogy for a real computer. It is a good analogy since it is still largely valid 50 years later.

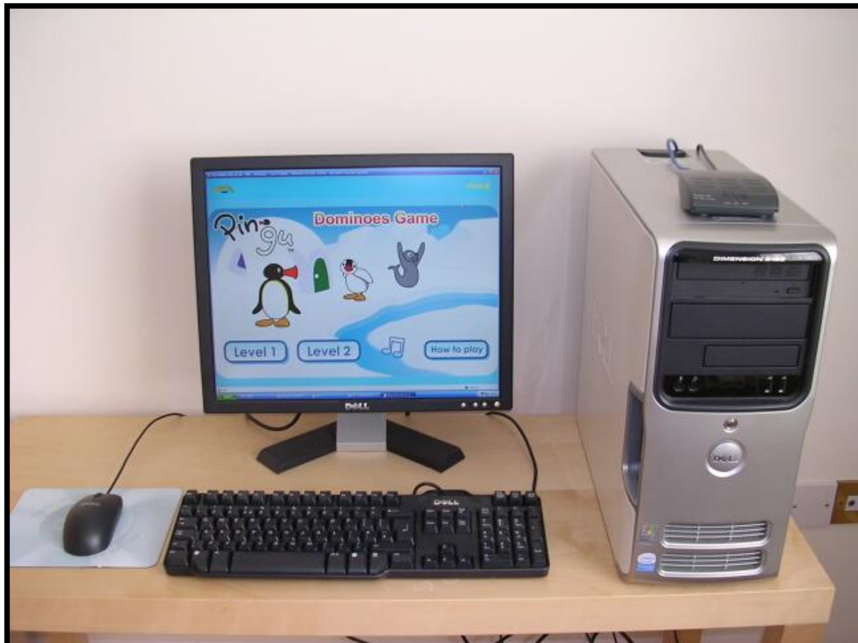
- CPU (Central Processing Unit)
- RAM (Random Access Memory ... Main Memory)
- Buses & Input/Output
- Assembly Language
- Machine Code

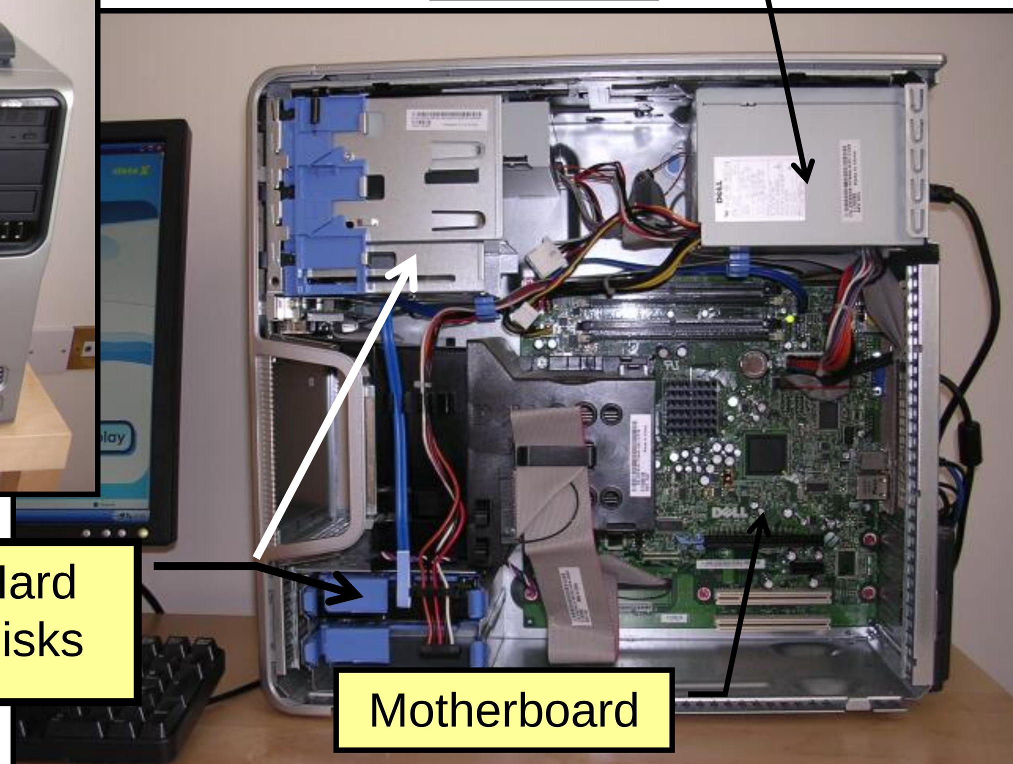
The LMC computer follows this “stored program” von Neumann Architecture ...



But is this really like a modern computer?

Real hardware ... my first Dell PC …
(“Dell Boy” now lives in my attic)





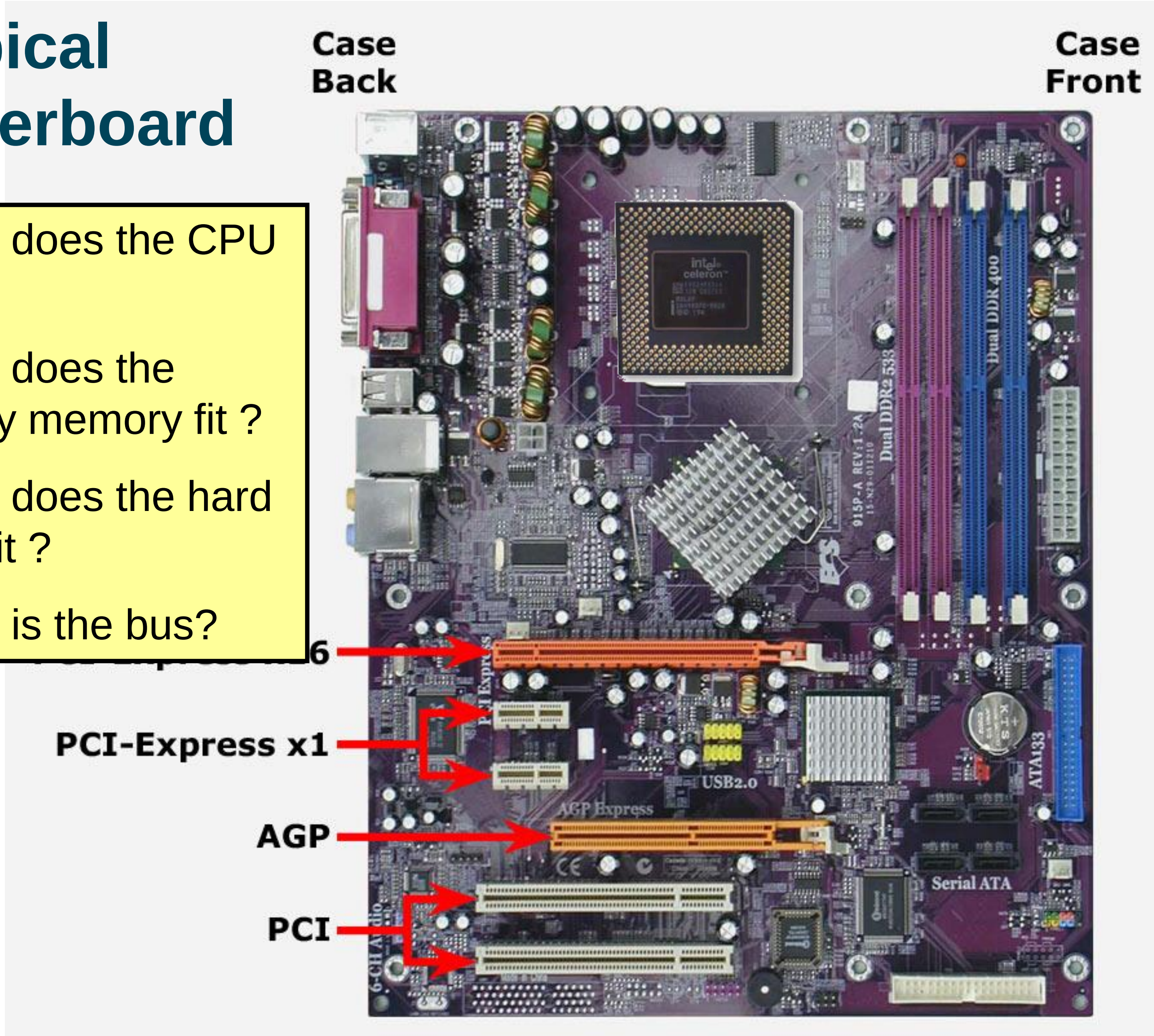
A typical Motherboard

Where does the CPU go ?

Where does the primary memory fit ?

Where does the hard disks fit ?

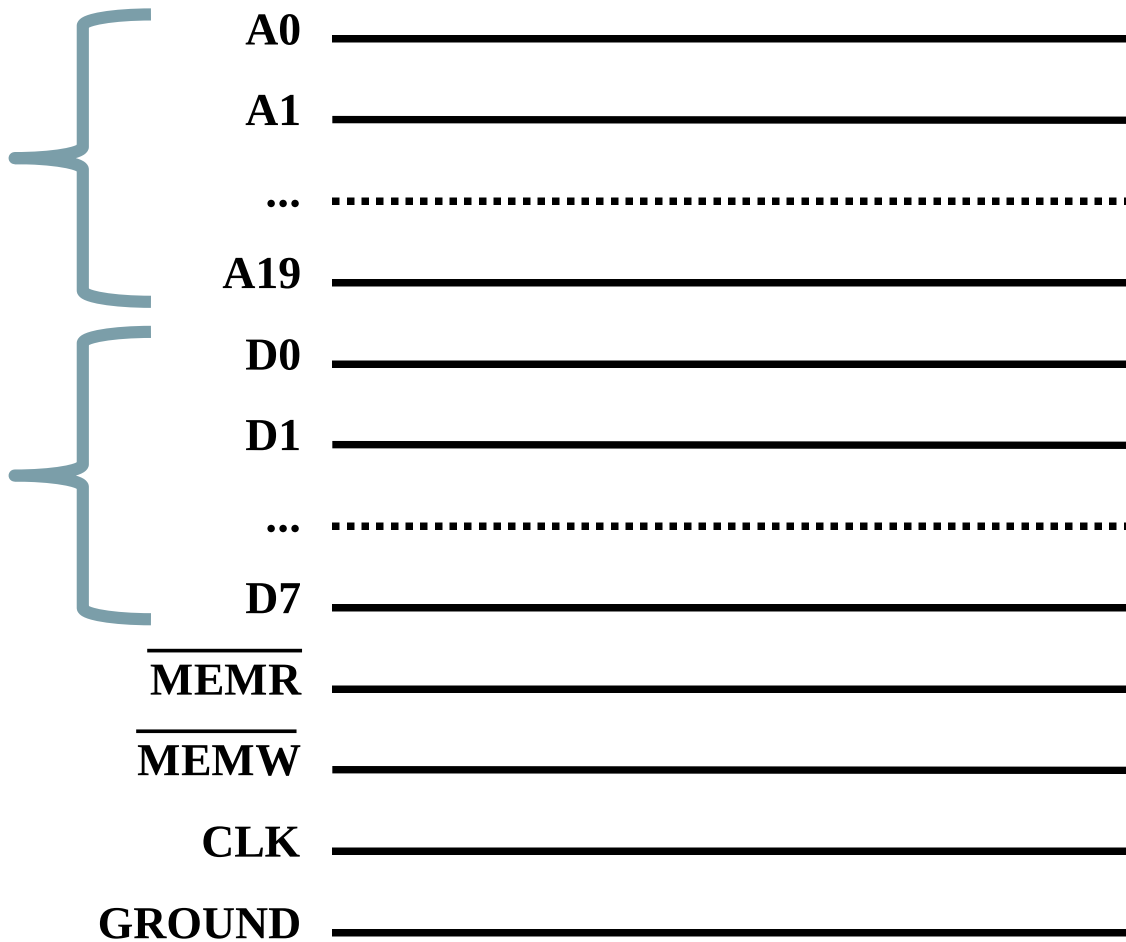
Where is the bus?



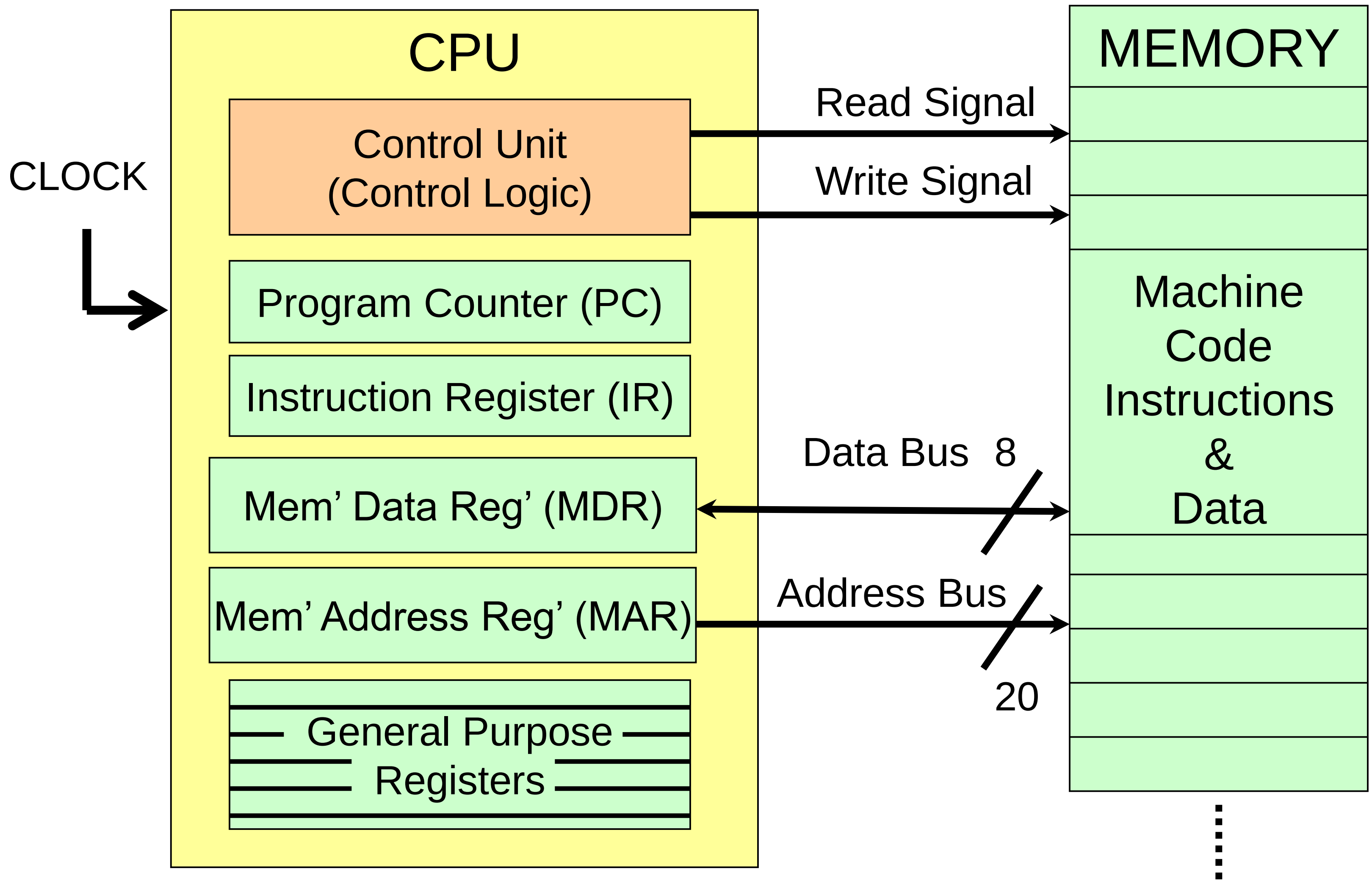
Original IBM PC External Bus just consists of lots of wires that carry binary signals ...

0 volts = “0” = off and **+5 volts = “1” = on**

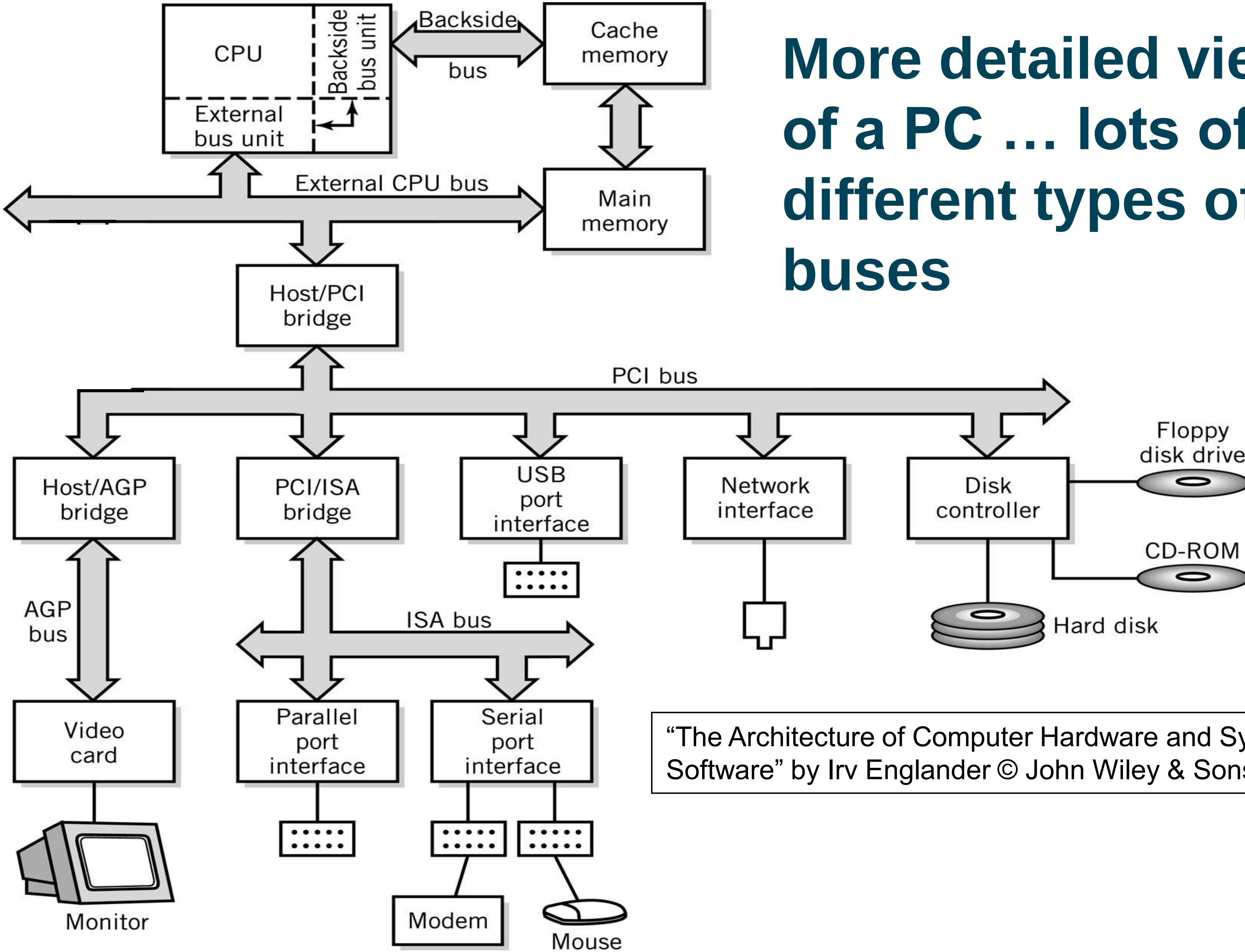
- 20 bit “**address lines**” can address 2^20 = 1MB of locations.
- 8 bit “**data lines**” can send a byte of data at a time.
- Control lines include read/write, clock and ground.



Interaction between CPU and Memory



More detailed view of a PC … lots of different types of buses



"The Architecture of Computer Hardware and Systems Software" by Irv Englander © John Wiley & Sons

A processor's (or programmer's) abstract view of the computer architecture …